

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 2 HOURS

SUMMER SEMESTER, 2023-2024
FULL MARKS: 120

CSE 4205: Digital Logic Design

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. Assume that Excess-5 is a coding scheme where a decimal number is represented by adding $(5)_{10}$ to its binary equivalent. For example, $(0)_{10}$ is represented as $(0101)_{XS-5}$ and decimal $(9)_{10}$ is $(1110)_{XS-5}$.

The Excess-5 encoder accepts only valid codes representing decimal digits from $(0)_{10}$ to $(9)_{10}$. All other input combinations are considered invalid and their corresponding outputs are treated as don't care conditions.

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|---|----------------------|
| a) Draw the truth table for the Binary-to-Excess-5 encoder based on the information provided. Clearly label all input and output columns, and appropriately handle the specified conditions for invalid inputs. | 4
(CO1)
(PO1) |
| b) Derive the simplified expressions for each of the output bits using K-map. | 16
(CO3)
(PO2) |
| c) Construct a two-level logic circuit for each output bit using only NAND gates. Use NAND gates to generate complemented input variables as needed (this preprocessing is not considered part of the two-level implementation). Draw the logic circuits for all output bits together in a single diagram, using the same set of input variables. | 7
(CO2)
(PO2) |
| d) Define what a self-complementing code is. Excess-3 is known to be a self-complementing code, but is Excess-5 also a self-complementing code? If yes, provide an example to illustrate why. If not, present a counterexample to demonstrate why it does not satisfy the property. | 3
(CO1)
(PO1) |

2. a) In digital communication, a checksum is a simple error-detection method used to verify data integrity. The sender transmits data along with a checksum (typically the sum of the values). The receiver then adds the received data and compares the result to the checksum. If the difference is zero, the data is assumed to be unaltered.

Alice intends to send five numbers to Bob, along with their checksum. However, due to transmission loss, only four of the numbers and the checksum reach him. Although a checksum is typically used for error detection rather than error correction or recovery, Bob assumes that the received numbers and the checksum remained intact during transmission and attempts to use them to recover the missing number.

The four numbers and the checksum that reached Bob are:

$$numbers = (785)_{10}, (102)_{10}, (637)_{10}, (954)_{10}$$

$$checksum = (10630)_7$$

- | | |
|--|---------------------|
| i. Convert the four numbers into their corresponding Binary-Coded Decimal (BCD) representations. | 4
(CO1)
(PO1) |
|--|---------------------|

- ii. Find the sum of the four numbers obtained in Question 2.a)i using Binary-Coded Decimal (BCD) addition. 8 (CO2)
- iii. Convert the sum obtained in Question 2.a)ii into a base 7 number and subtract it from the checksum using Radix complement (7's complement) to compute the number that was lost during transmission (use a 6-digit register, where the most significant bit (MSB) or the leftmost digit, denotes the sign bit). 2 + 8 (PO2) (CO1) (PO1)
- b) Can the checksum result be zero even if the transmitted numbers have been modified? In other words, can a change go undetected when using a checksum? Briefly explain your reasoning with an example. 5 (CO1) (PO1)
- c) Even though binary and BCD representations of decimal digits are quite similar, why does BCD addition require an adjustment step that binary addition does not? 3 (CO1) (PO1)
3. a) Simplify the following boolean expression using Quine-McCluskey (Tabular) method. 15 (CO5) (PO1)
- $$F(A, B, C, D) = \prod M(3, 5, 6, 9, 12) \cdot \prod d(2, 10, 14)$$
- While solving using the Tabular Method, clearly state the prime implicants and essential prime implicants.
- b) Using cut and try method, prove the following - 5 + 5 (CO2) (PO2)
- i. $xy + x'z + yz = xy + x'z$
- ii. $xy'z + x'yz + xy = xy + yz + zx$
- c) Can parity always successfully detect an error? If yes, briefly justify your answer, if not, provide a counterexample. 2 + 3 (CO1) (PO1)
- A system uses odd parity for error detection. Which of the following bit patterns (including the parity bit) indicates an error?
- i. 1101010
- ii. 1011101
4. a) What is the advantage of using a lookahead carry adder compared to a ripple carry adder? Provide the necessary equations for carry generation and propagation to support your explanation. Also, compare their circuit or block diagrams to illustrate the difference. 5 + 10 (CO5) (PO1)
- b) Design a 3-bit Full Adder-Subtractor circuit and explain its working mechanism, including how it functions as both an adder and a subtractor. 5 + 5 (CO5) (PO3)
- c) Is the circuit shown in Figure 1 a sequential or combinational circuit? Justify your answer. 5 (CO1) (PO1)

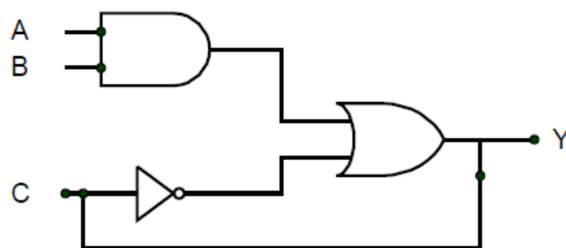


Figure 1: A boolean circuit